# Rational Functions & Asymptotes

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.

Factor: $f(x) = \frac{(x-3)(x+3)}{(x-4)(x+3)}$. The factor $(x + 3)$ cancels, giving a hole at $x = -3$. The remaining denominator factor $(x - 4)$ gives a vertical asymptote at $x = 4$.

**2** **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the $y$-coordinate of the hole in the function from the previous problem.

Simplified form (away from the hole): $f(x) = \frac{x-3}{x-4}$. At $x = -3$: $f(-3) = \frac{-3-3}{-3-4} = \frac{-6}{-7} = \frac{6}{7}$. Hole at $\left(-3, \frac{6}{7}\right)$.

**3** **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{3}{5}$.

**4** **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$.

**5** **[Foundational]** • **Target: Horizontal and slant asymptotes**

Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

The numerator's degree (2) is exactly one more than the denominator's degree (1), producing a slant asymptote. Dividing: $x^2 + 1 = (x - 2)(x + 2) + 5$, so $h(x) = x + 2 + \frac{5}{x-2}$. As $x \to \pm\infty$, the remainder term vanishes, leaving the slant asymptote $y = x + 2$.

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**6** [Foundational] • Target: Visualizing a rational function's asymptotes

The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.

**a** Identify the vertical asymptote shown. Vertical asymptote at $x = 2$, where the two branches shoot off toward $\pm\infty$.

**b** Identify the horizontal asymptote shown. Horizontal asymptote at $y = 1$, which both branches approach far from $x = 2$ (this is the reciprocal function $\frac{1}{x}$ shifted right 2 and up 1).

(a) Vertical asymptote at $x = 2$, where the two branches shoot off toward $\pm\infty$.; (b) Horizontal asymptote at $y = 1$, which both branches approach far from $x = 2$ (this is the reciprocal function $\frac{1}{x}$ shifted right 2 and up 1).

**7** [Foundational] • Target: End behavior of rational functions

Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \to \pm\infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).

Since the degree of the numerator (3) exceeds the denominator (2) by exactly 1, $f$ has a slant asymptote and $f(x) \to \pm\infty$ as $x \to \pm\infty$, following the line found by dividing $2x^3 - x$ by $x^2 + 1$, approximately $y = 2x$ for large $|x|$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Domain and intercepts

Find the domain, $x$-intercept(s), and $y$-intercept of $f(x) = \frac{x-5}{x^2-4}$.

Domain: all reals except $x = \pm 2$ (where the denominator is zero), so $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$. $x$-intercept: set numerator to 0: $x = 5$, giving $(5, 0)$. $y$-intercept: $f(0) = \frac{-5}{-4} = \frac{5}{4}$, giving $\left(0, \frac{5}{4}\right)$.

**9** [Intermediate] • Target: Solving rational equations and inequalities

Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.

$x + 1 = 2(x - 3) = 2x - 6$, so $-x = -7$, $x = 7$. Check: $x - 3 = 4 \neq 0$, valid. Solution: $x = 7$.

**10** [Intermediate] • Target: Solving rational equations and inequalities

Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.

Critical points: $x = 1$ (zero) and $x = -2$ (excluded). Testing intervals: for $x < -2$, both factors negative, ratio positive. For $-2 < x < 1$, numerator negative, denominator positive, ratio negative. For $x > 1$, both positive, ratio positive. Solution: $x < -2$ or $x > 1$, i.e. $(-\infty, -2) \cup (1, \infty)$.

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**11** **[Intermediate]** • **Target: Visualizing a function with a slant asymptote**

The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.

**a** Find the vertical asymptote and confirm it matches the graph. Vertical asymptote at $x = 1$, matching the break in the graph shown there.

**b** Use polynomial division to find the slant asymptote equation. Dividing: $x^2 = (x-1)(x+1) + 1$, so $f(x) = x + 1 + \frac{1}{x-1}$. Slant asymptote: $y = x + 1$, matching the line the branches hug for large $|x|$.

**c** Explain why this function has no horizontal asymptote. Since the numerator's degree (2) exceeds the denominator's degree (1), the function grows without bound rather than leveling off, so there is no horizontal asymptote — only the slant asymptote found above.

(a) Vertical asymptote at $x = 1$, matching the break in the graph shown there.; (b) Dividing: $x^2 = (x-1)(x+1) + 1$, so $f(x) = x + 1 + \frac{1}{x-1}$. Slant asymptote: $y = x + 1$, matching the line the branches hug for large $|x|$.; (c) Since the numerator's degree (2) exceeds the denominator's degree (1), the function grows without bound rather than leveling off, so there is no horizontal asymptote — only the slant asymptote found above.

**12** **[Intermediate]** • **Target: Building a rational function from given features**

Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

Start with $\frac{2(x+1)}{x+1} \cdot \frac{1}{x-3}$-style construction: use $f(x) = \frac{2(x+1)^2}{(x+1)(x-3)}$. The common factor $(x+1)$ creates the hole at $x = -1$; the remaining $(x-3)$ in the denominator gives the vertical asymptote; matching leading coefficients (2 over 1) gives horizontal asymptote $y = 2$.

**13** **[Intermediate]** • **Target: Building a rational function from given features**

Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

A horizontal asymptote occurs when the numerator's degree is less than or equal to the denominator's degree, while a slant asymptote requires the numerator's degree to be exactly one more than the denominator's. These are mutually exclusive degree conditions, so a rational function has at most one of the two (or neither, if the numerator's degree exceeds the denominator's by more than 1).

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**14** **[Intermediate]** • **Target: Rational functions in context**

The concentration of a drug in the bloodstream $t$ hours after a dose is modeled by $C(t) = \frac{50t}{t^2+4}$ (mg/L). Find the horizontal asymptote and interpret it in context.

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$. In context, this means the drug concentration approaches 0 mg/L as time goes on — the drug is eventually fully eliminated from the bloodstream.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Horizontal and slant asymptotes** • **Cross-Lesson: 4.01 Inverse & Composite Functions**

Let $f(x) = \frac{4e^{2x}}{e^{2x}+3}$. Prove that $f$ is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of $f^{-1}$.

Step 1: Solve $y = \frac{4e^{2x}}{e^{2x}+3}$ for $e^{2x}$: $y(e^{2x}+3) = 4e^{2x} \Rightarrow 3y = e^{2x}(4-y) \Rightarrow e^{2x} = \frac{3y}{4-y}$. Step 2: Take the natural logarithm and divide by 2: $2x = \ln\left(\frac{3y}{4-y}\right) \Rightarrow f^{-1}(x) = \mathbf{\frac{1}{2}}\ln\left(\mathbf{\frac{3x}{4-x}}\right)$. Step 3: Domain of $f^{-1}$ (range of $f$): Since $e^{2x} > 0$, we require $\frac{3x}{4-x} > 0 \Rightarrow \mathbf{0 < x < 4}$.

**16** **[Challenging]** • **Target: Visualizing a rational function's asymptotes** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Let $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x+5}$. Find the composite function $(f \circ g)(x)$ and state its exact domain.

Step 1: Determine the domain of the inner function $g(x)$: $x + 5 \geq 0 \Rightarrow x \geq -5$. Step 2: Substitute $g(x)$ into $f(x)$: $(f \circ g)(x) = f\left(\sqrt{x+5}\right) = 2\left(\sqrt{x+5}\right)^2 - 3 = 2(x+5) - 3 = \mathbf{2x+7}$, with domain $\mathbf{[-5, \infty)}$.

**17** **[Challenging]** • **Target: End behavior of rational functions** • **Cross-Lesson: 4.03 Parametric Equations & Modeling**

For a quadratic function $f(x)$, selected values are $f(1) = 4$, $f(2) = 9$, $f(3) = 18$, and $f(4) = 31$. Compute the first and second differences, verify that $f$ is quadratic, and find its explicit formula $f(x) = ax^2 + bx + c$.

Step 1: Compute first and second differences:First differences $\Delta y$: $9 - 4 = 5$, $18 - 9 = 9$, $31 - 18 = 13$ .Second differences $\Delta^2 y$: $9 - 5 = 4$, $13 - 9 = 4$.Since $\Delta^2 y = 4$ is constant, $f(x)$ is quadratic with $2a = 4 \Rightarrow a = 2$. Step 2: Solve for $b$ and $c$:$f(1) = 2(1)^2 + b(1) + c = 4 \Rightarrow b + c = 2$. $f(2) = 2(2)^2 + b(2) + c = 9 \Rightarrow 2b + c = 1$.Subtracting gives $b = -1$ and $c = 3$. Thus $\mathbf{f(x) = 2x^2 - x + 3}$ .

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**18** [Challenging] • Target: Domain and intercepts • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A bivariate dataset $(x, y)$ is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y}) = 1.5 + 0.4x$. Express $\hat{y}$ as an exponential function of $x$ in the form $\hat{y} = a \cdot b^x$, and find the predicted percentage growth rate per unit increase in $x$.

Step 1: Exponentiate both sides of the semi-log regression equation: $\hat{y} = e^{1.5 + 0.4x} = e^{1.5} \cdot (e^{0.4})^x$. Step 2: Evaluate parameters $a$ and $b$:
$a = e^{1.5} \approx 4.4817, \quad b = e^{0.4} \approx 1.4918 \Rightarrow \mathbf{\hat{y} = e^{1.5}(e^{0.4})^x \approx 4.482(1.492)^x}$. Step 3: Percentage growth rate: Since $b - 1 = e^{0.4} - 1 \approx 0.4918$, $\hat{y}$ increases by approximately **49.18%** per unit increase in $x$.

**19** [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 1.03 Rates of Change & Concavity

Prove the trigonometric identity $\frac{\sin(2x)}{1 + \cos(2x)} = \tan x$ and state the domain exclusions.

Step 1: Apply double-angle identities $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = 2\cos^2 x - 1$:
LHS $= \frac{2\sin x\cos x}{1 + (2\cos^2 x - 1)} = \frac{2\sin x\cos x}{2\cos^2 x}$. Step 2: Cancel $2\cos x$ (for $\cos x \neq 0$): $\frac{\sin x}{\cos x} = \tan x =$ RHS. ■ Domain exclusions: $\mathbf{x \neq \frac{\pi}{2} + k\pi}$ for $k \in \mathbb{Z}$.

**20** [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

Step 1: Group and complete the square: $4(x^2 - 4x + 4) - 9(y^2 + 2y + 1) = 29 + 16 - 9 = 36$.
$\mathbf{\frac{(x-2)^2}{9} - \frac{(y+1)^2}{4} = 1}$. Step 2: Extract hyperbola properties ($a = 3, b = 2, c = \sqrt{9 + 4} = \sqrt{13}$):Center: $(2, -1)$Vertices: $(-1, -1)$ and $(5, -1)$Foci: $(2 \pm \sqrt{13}, -1)$Asymptotes: $y + 1 = \pm\frac{2}{3}(x - 2)$

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Vertical asymptotes and holes

Find the horizontal asymptote of $f(x) = \frac{9x^2 + 5x - 2}{6x^2 + x - 7}$.

**a.** $y = \frac{2}{3}$

**b.** $y = \frac{5}{2}$

**c.** $y = 0$

**d.** $y = \frac{3}{2}$ ✓

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{9}{6} = \frac{3}{2}$.

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**22** [Foundational] • Target: Horizontal and slant asymptotes

Find the horizontal asymptote of $f(x) = \frac{2x^2 - 9x - 9}{2x^2 - 6x - 3}$.

**a.** $y = \frac{3}{2}$

**b.** $y = 0$

**c.** $y = 1$ ✓

**d.** $y = 2$

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{2} = 1$.

**23** [Foundational] • Target: Visualizing a rational function's asymptotes

Find the horizontal asymptote of $f(x) = \frac{4x^2 - 8x + 6}{9x^2 + 3x + 9}$.

**a.** $y = \frac{9}{4}$

**b.** $y = \frac{4}{9}$ ✓

**c.** $y = 0$

**d.** $y = \frac{13}{9}$

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{4}{9} = \frac{4}{9}$.

**24** [Foundational] • Target: End behavior of rational functions

Find the horizontal asymptote of $f(x) = \frac{6x^2 - 7x}{4x^2 - 7x - 9}$.

**a.** $y = \frac{3}{2}$ ✓

**b.** $y = \frac{2}{3}$

**c.** $y = 0$

**d.** $y = \frac{5}{2}$

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{6}{4} = \frac{3}{2}$.

**25** [Foundational] • Target: Domain and intercepts

Find the horizontal asymptote of $g(x) = \frac{9x + 3}{7x^2 + 7x + 5}$.

**a.** $y = 0$ ✓

**b.** $y = 1$

**c.** $y = \frac{9}{7}$

**d.** $y = -\frac{9}{7}$

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$.

**26** [Foundational] • Target: Solving rational equations and inequalities

Find the horizontal asymptote of $f(x) = \frac{9x^2 + 6x - 3}{8x^2 + 3x - 5}$.

**a.** $y = 0$

**b.** $y = \frac{9}{8}$ ✓

**c.** $y = \frac{8}{9}$

**d.** $y = \frac{17}{8}$

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{9}{8} = \frac{9}{8}$.

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**27** [Foundational] • Target: Visualizing a function with a slant asymptote

Find the horizontal asymptote of $g(x) = \frac{-7x+1}{4x^2-4x+4}$.

**a.** $y = \frac{7}{4}$

**b.** $y = 0$ ✓

**c.** $y = -\frac{7}{4}$

**d.** $y = 1$

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$.

**28** [Foundational] • Target: Building a rational function from given features

Find the hole and vertical asymptote of $f(x) = \frac{(x+2)(x-5)}{(x+2)(x-6)}$.

**a.** Hole at $x = 6$; vertical asymptote at $x = -2$.

**b.** **Hole at $x = -2$; vertical asymptote at $x = 6$.** ✓

**c.** Vertical asymptotes at $x = -2$ and $x = 6$, no hole.

**d.** Hole at $x = 5$; vertical asymptote at $x = 6$.

The factor $(x + 2)$ cancels, giving a hole at $x = -2$. The remaining denominator factor $(x - 6)$ gives a vertical asymptote at $x = 6$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Rational functions in context

Find the hole and vertical asymptote of $f(x) = \frac{(x-2)(x-4)}{(x-2)(x+6)}$.

**a.** **Hole at $x = 2$; vertical asymptote at $x = -6$.** ✓

**b.** Vertical asymptotes at $x = 2$ and $x = -6$, no hole.

**c.** Hole at $x = -6$; vertical asymptote at $x = 2$.

**d.** Hole at $x = 4$; vertical asymptote at $x = -6$.

The factor $(x - 2)$ cancels, giving a hole at $x = 2$. The remaining denominator factor $(x + 6)$ gives a vertical asymptote at $x = -6$.

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**30** [Intermediate] • Target: Vertical Asymptotes vs. Removable Discontinuities

Find the hole and vertical asymptote of $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-3)}$.

**a.** Vertical asymptotes at $x = -8$ and $x = 3$, no hole.

**b.** **Hole at $x = -8$; vertical asymptote at $x = 3$.** ✓

**c.** Hole at $x = 8$; vertical asymptote at $x = 3$.

**d.** Hole at $x = 3$; vertical asymptote at $x = -8$.

The factor $(x + 8)$ cancels, giving a hole at $x = -8$. The remaining denominator factor $(x - 3)$ gives a vertical asymptote at $x = 3$.

**31** [Intermediate] • Target: Horizontal Asymptotes and Degrees

Find the hole and vertical asymptote of $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-9)}$.

**a.** **Hole at $x = -8$; vertical asymptote at $x = 9$.** ✓

**b.** Vertical asymptotes at $x = -8$ and $x = 9$, no hole.

**c.** Hole at $x = 8$; vertical asymptote at $x = 9$.

**d.** Hole at $x = 9$; vertical asymptote at $x = -8$.

The factor $(x + 8)$ cancels, giving a hole at $x = -8$. The remaining denominator factor $(x - 9)$ gives a vertical asymptote at $x = 9$.

**32** [Intermediate] • Target: Slant (Oblique) Asymptotes

Find the hole and vertical asymptote of $f(x) = \frac{(x-1)(x+1)}{(x-1)(x)}$.

**a.** **Hole at $x = 1$; vertical asymptote at $x = 0$.** ✓

**b.** Vertical asymptotes at $x = 1$ and $x = 0$, no hole.

**c.** Hole at $x = 0$; vertical asymptote at $x = 1$.

**d.** Hole at $x = -1$; vertical asymptote at $x = 0$.

The factor $(x - 1)$ cancels, giving a hole at $x = 1$. The remaining denominator factor $(x)$ gives a vertical asymptote at $x = 0$.

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**33** [Intermediate] • Target: Limit Notation and Asymptotic Behavior

Find the hole and vertical asymptote of $f(x) = \frac{(x-6)(x+1)}{(x-6)(x-5)}$.

**a.** Hole at $x = -1$; vertical asymptote at $x = 5$.

**b.** Vertical asymptotes at $x = 6$ and $x = 5$, no hole.

**c.** **Hole at $x = 6$; vertical asymptote at $x = 5$.** ✓

**d.** Hole at $x = 5$; vertical asymptote at $x = 6$.

The factor $(x - 6)$ cancels, giving a hole at $x = 6$. The remaining denominator factor $(x - 5)$ gives a vertical asymptote at $x = 5$.

**34** [Intermediate] • Target: Vertical asymptotes and holes

Find the hole and vertical asymptote of $f(x) = \frac{(x+6)(x+4)}{(x+6)(x+7)}$.

**a.** Vertical asymptotes at $x = -6$ and $x = -7$, no hole.

**b.** Hole at $x = -4$; vertical asymptote at $x = -7$.

**c.** Hole at $x = -7$; vertical asymptote at $x = -6$.

**d.** **Hole at $x = -6$; vertical asymptote at $x = -7$.** ✓

The factor $(x + 6)$ cancels, giving a hole at $x = -6$. The remaining denominator factor $(x + 7)$ gives a vertical asymptote at $x = -7$.

**35** [Intermediate] • Target: Horizontal and slant asymptotes

Find the slant asymptote of $h(x) = \frac{x^2 + 8x - 7}{x + 9}$.

**a.** $y = x$

**b.** $y = x - 2$

**c.** $y = x + 1$

**d.** $y = x - 1$ ✓

Dividing $x^2 + 8x - 7$ by $x + 9$: quotient $= x - 1$, with a remainder that vanishes as $x \to \pm\infty$. Slant asymptote: $y = x - 1$.

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**36** [Intermediate] • Target: Visualizing a rational function's asymptotes

Find the slant asymptote of $h(x) = \frac{x^2+4x+5}{x+6}$.

**a.** $y = x - 3$

**b.** $y = x - 1$

**c.** $y = x - 2$ ✓

**d.** $y = x + 2$

Dividing $x^2 + 4x + 5$ by $x + 6$: quotient $= x - 2$, with a remainder that vanishes as $x \to \pm\infty$. Slant asymptote: $y = x - 2$.

**37** [Intermediate] • Target: End behavior of rational functions

Solve $\frac{x+4}{x-3} = -3$ for $x$.

**a.** $-\frac{5}{4}$

**b.** $\frac{5}{4}$ ✓

**c.** $-\frac{4}{3}$

**d.** $\frac{9}{4}$

$x + 4 = -3(x - 3)$. Solving: $(4)x = 5$, so $x = \frac{5}{4}$ (denominator $x - 3 \neq 0$ checked).

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Domain and intercepts • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve $\frac{x+3}{x} = -2$ for $x$.

**a.** $1$

**b.** $-\frac{3}{2}$

**c.** $0$

**d.** $-1$ ✓

$x + 3 = -2(x)$. Solving: $(3)x = -3$, so $x = -1$ (denominator $x \neq 0$ checked).

**39** [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 1.03 Rates of Change & Concavity

Solve the inequality $\frac{x-6}{x-7} > 0$.

**a.** $(-\infty, 6) \cup (7, \infty)$ ✓

**b.** $[6, 7]$

**c.** $(6, 7)$

**d.** $(-\infty, 7) \cup (6, \infty)$

Critical points: $x = 6$ (zero) and $x = 7$ (excluded). The expression has the same sign as $(x - 6)(x - 7)$, which is positive outside $[6, 7]$. Solution: $(-\infty, 6) \cup (7, \infty)$.

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**40** [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Solve the inequality $\frac{x-2}{x-7} > 0$.

**a.** $(-\infty, 7) \cup (2, \infty)$

**b.** $(2, 7)$

**c.** $(-\infty, 2) \cup (7, \infty)$ ✓

**d.** $[2, 7]$

Critical points: $x = 2$ (zero) and $x = 7$ (excluded). The expression has the same sign as $(x - 2)(x - 7)$, which is positive outside $[2, 7]$. Solution: $(-\infty, 2) \cup (7, \infty)$.

**41** [Challenging] • Target: Vertical Asymptotes vs. Removable Discontinuities • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 9}{x^2 - x - 6}$.

**a.** **Vertical asymptote at $x = -2$; hole at $x = 3$** ✓

**b.** Vertical asymptote at $x = 3$; hole at $x = -2$

**c.** Vertical asymptotes at $x = 3$ and $x = -2$; no holes

**d.** Holes at $x = 3$ and $x = -2$; no vertical asymptotes

Factor: $\frac{(x-3)(x+3)}{(x-3)(x+2)}$. The factor $(x - 3)$ cancels, indicating a removable discontinuity (hole) at $x = 3$. The remaining denominator factor $(x + 2)$ yields a vertical asymptote at $x = -2$.

**42** [Challenging] • Target: Vertical Asymptotes vs. Removable Discontinuities • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

If $h(x) = \frac{x-4}{(x-4)^2(x+1)}$, which describes the behavior of $h$ near $x = 4$?

**a.** $h(x)$ **has a vertical asymptote at** $x = 4$ ✓

**b.** $h(x)$ has a removable hole at $x = 4$

**c.** $h(x)$ has a horizontal tangent at $x = 4$

**d.** $h(x)$ is continuous at $x = 4$

Simplifying: $h(x) = \frac{1}{(x-4)(x+1)}$ for $x \neq 4$. Because $(x - 4)$ remains in the denominator to degree 1, $x = 4$ is a vertical asymptote, not a hole.

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**43** [Challenging] • Target: Horizontal Asymptotes and Degrees • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Can the graph of a rational function intersect its horizontal asymptote?

**a. Yes, the graph can cross a horizontal asymptote at finite values of $x$ ✓**

**b.** No, an asymptote can never be touched or crossed by definition

**c.** Only if the horizontal asymptote is $y = 0$

**d.** Only if the function has no vertical asymptotes

Horizontal asymptotes describe end behavior as $x \to \pm\infty$. The graph of a rational function can intersect a horizontal asymptote one or more times in the interior/finite domain.

**44** [Challenging] • Target: Slant (Oblique) Asymptotes • Cross-Lesson: 4.01 Inverse & Composite Functions

Under what condition does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

**a. The degree of $P(x)$ is exactly 1 greater than the degree of $Q(x)$ ✓**

**b.** The degree of $P(x)$ equals the degree of $Q(x)$

**c.** The degree of $Q(x)$ is exactly 1 greater than the degree of $P(x)$

**d.** The degree of $P(x)$ is at least 2 greater than the degree of $Q(x)$

A slant asymptote exists if and only if the numerator's polynomial degree is exactly 1 higher than the denominator's polynomial degree.

**45** [Challenging] • Target: Limit Notation and Asymptotic Behavior • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

For $f(x) = \frac{1}{x-3}$, what are $\lim_{x\to 3^+} f(x)$ and $\lim_{x\to 3^-} f(x)$?

**a. $\lim_{x\to 3^+} f(x) = \infty$ and $\lim_{x\to 3^-} f(x) = -\infty$ ✓**

**b.** $\lim_{x\to 3^+} f(x) = -\infty$ and $\lim_{x\to 3^-} f(x) = \infty$

**c.** Both limits equal $\infty$

**d.** Both limits equal $-\infty$

As $x \to 3$ from the right, $x - 3 > 0$, so $\frac{1}{x-3} \to +\infty$. As $x \to 3$ from the left, $x - 3 < 0$, so $\frac{1}{x-3} \to -\infty$.

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